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Doxycycline Postexposure Prophylaxis and Sexually Transmitted Infection Trends

Madeline Sankaran, MPH; David V. Glidden, PhD; Robert P. Kohn, MPH; Trang Q. Nguyen, PhD; Oliver Bacon, MD; Susan P. Buchbinder, MD; Monica Gandhi, MD; Diane V. Havlir, MD; Courtney Liebi, MPP; Anne F. Luetkemeyer, MD; Janet Q. Nguyen, MPH; Jorge Roman, MSN; Hyman Scott, MD; Thiago S. Torres, PhD; Stephanie E. Cohen, MD

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IMPORTANCE Increasing rates of sexually transmitted infections (STIs) have been associated with rises in serious morbidity. While doxycycline postexposure prophylaxis (doxyPEP), a strategy in which individuals take doxycycline, 200 mg, after condomless sex to prevent bacterial STIs, has been shown to be efficacious in randomized clinical trials, doxyPEP's potential effect on population-level STI incidence is unknown.

OBJECTIVE To assess the association of citywide doxyPEP guideline release with reported chlamydia, gonorrhea, and early syphilis cases in men who have sex with men (MSM) and in transgender women in San Francisco, California.

DESIGN, SETTING, AND PARTICIPANTS This population-level interrupted time series analysis of reported San Francisco STI cases measured monthly cases of chlamydia, gonorrhea, and early syphilis prior to (July 2021-October 2022) and after (November 2022-November 2023) release of citywide doxyPEP guidelines in October 2022. All reported chlamydia, gonorrhea, and early syphilis cases among MSM and transgender women in San Francisco during the period of analysis were included. Data were analyzed November 2023 to July 2024.

EXPOSURE Release of doxyPEP citywide guidelines.

MAIN OUTCOMES AND MEASURES The primary outcome was the percentage change between projected and observed chlamydia, gonorrhea, and early syphilis cases in the 13-month postexposure period.

RESULTS Citywide, there were 6694 cases of chlamydia, 9603 cases of gonorrhea, and 2121 cases of early syphilis among MSM and transgender women during the analytic period. STI cases among MSM and transgender women decreased significantly compared with model projections for chlamydia (−6.58% per month; 95% CI, −7.99% to −5.16%) and early syphilis (−2.68% per month; 95% CI, −3.75% to −1.60%) after doxyPEP implementation. By the end of the 13-month postperiod in November 2023, chlamydia and early syphilis cases decreased −49.64% (95% CI, −59.05% to −38.06%) and −51.39% (95% CI, −58.21% to −43.46%), respectively, compared with projected cases. There was a significant increase in monthly gonorrhea cases compared with projections (1.77% per month; 95% CI, 0.87% to 2.67%).

CONCLUSIONS AND RELEVANCE This study suggests that San Francisco's doxyPEP guideline release was associated with decreases in reported cases of chlamydia and early syphilis, but not gonorrhea, among MSM and transgender women in San Francisco. Further analyses are needed to assess whether declines are sustained and monitor for adverse consequences, including antimicrobial resistance. Supporting doxyPEP implementation for MSM and transgender women at risk for STIs could have a significant impact on the nationwide STI epidemic.

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Author Affiliations: San Francisco Department of Public Health, San Francisco, California (Sankaran, Kohn, T. Q. Nguyen, Bacon, Buchbinder, Scott, Cohen); University of California, San Francisco (Glidden, Bacon, Buchbinder, Gandhi, Havlir, Liebi, Luetkemeyer, J. Q. Nguyen, Scott, Cohen); San Francisco AIDS Foundation, San Francisco, California (Roman, Scott); Instituto Nacional de Infectologia Evandro Chagas, Fundação Oswaldo Cruz, Rio de Janeiro, Brazil (Torres).

Corresponding Author: Stephanie E. Cohen, MD, San Francisco Department of Public Health, 25 Van Ness Ave, San Francisco, CA 94102 (stephanie.cohen@sfdph.org).

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Rates of bacterial sexually transmitted infections (STIs) have increased in the US over the past decade.[1] Increasing rates of STIs have been associated with a rise in serious morbidity, including congenital syphilis and ocular syphilis.[1,2] STIs increase the risk of HIV transmission and acquisition. Gay, bisexual, and other cisgender men who have sex with men (MSM), transgender women, and some racial and ethnic minority groups, including African American/Black persons and Hispanic/Latinx persons, have disproportionately higher STI rates.[1] San Francisco, California, a city of nearly 880 000 persons, including an estimated 23 000 to 66 000 MSM, has the highest rates of chlamydia, gonorrhea, and early syphilis of any county in California and one of the highest rates of STIs of any jurisdiction in the US.[1,3-5] From 2014 to 2019, reported cases of chlamydia, gonorrhea, and early syphilis among MSM and transgender women in San Francisco increased 76%, 72%, and 45%, respectively (eFigure in Supplement 1). While reported cases decreased for all 3 infections during 2020 due to the COVID-19 pandemic, from 2020 to 2021 reported cases increased, although they did not return to pre-COVID-19 pandemic levels. As of 2021, there were 4133, 4548, and 1119 cases of chlamydia, gonorrhea, and early syphilis, respectively, among male individuals in San Francisco (926.4, 1019.4, and 250.8 cases per 100 000 population).[6] New tools for STI prevention are urgently needed to address rising rates of STIs and reduce STI incidence, morbidity, and disparities.

Doxycycline postexposure prophylaxis (doxyPEP) is a biomedical STI prevention strategy in which an individual takes doxycycline, 200 mg, as soon as possible within 72 hours after condomless oral, anal, or vaginal sex. Two randomized clinical trials, the DoxyPEP study in Seattle and San Francisco and the DOXYVAC study in France, demonstrated that doxyPEP reduced the risk of chlamydia, gonorrhea, and early syphilis among MSM and transgender women by approximately 65% overall and reduced incident chlamydia and early syphilis by approximately 80%.[7,8] DoxyPEP was less effective against incident gonorrhea, reducing it by 33% in the DOXYVAC study and 55% in the DoxyPEP study, which may be related to the prevalence of preexisting tetracycline-resistant *Neisseria gonorrhoeae* in the populations where it has been studied.[1,7-10] A study of doxyPEP in cisgender women in Kenya found no significant effect on bacterial STIs, but analyses of drug levels in hair suggested that doxycycline use in the study was low.[11,12]

Based on the results of these studies, in October 2022, the San Francisco Department of Public Health (SFDPH) became the first health jurisdiction in the US to develop and disseminate doxyPEP guidelines, recommending doxyPEP for MSM and transgender women with a history of an STI or multiple sex partners in the past year.[13] In June 2024, the US Centers for Disease Control and Prevention published doxyPEP guidelines that recommend that MSM and transgender women with a history of an STI in the past year be counseled on the use of doxyPEP for STI prevention as part of a comprehensive sexual health approach (Grade A1).[14] Due to insufficient evidence, most doxyPEP guidelines have not provided a recommendation for cisgender women.

While doxyPEP's efficacy has been demonstrated in randomized clinical trials, population-level impacts on STI incidence are unknown. In this analysis, we describe STI trends in San Francisco and assess whether there was a change in citywide reported cases of chlamydia, gonorrhea, and early syphilis in MSM and transgender women after the release and early implementation of citywide doxyPEP guidelines.

Key Points

Question Was implementation of doxycycline postexposure prophylaxis (doxyPEP) in San Francisco, California, associated with a decline in sexually transmitted infections (STIs) citywide?

Findings In this study including 6694 cases of chlamydia, 9603 cases of gonorrhea, and 2121 cases of early syphilis among men who have sex with men and in transgender women, there were significant, ongoing declines in reported cases of early syphilis and chlamydia in San Francisco after implementation of doxyPEP guidelines.

Meaning These findings suggest that doxyPEP could be an effective strategy for reducing population-level incidence of STIs.

Methods

This analysis was conducted by SFDPH using data routinely collected for public health surveillance according to California Code of Regulations, Title 17, §2500. The University of California, San Francisco institutional review board deemed it exempt from full review under Category 4 of the Revised Common Rule for which consent is not required. We followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Data Sources

To assess the ecologic association between doxyPEP guideline implementation and population-level, citywide STI incidence, we used citywide reportable disease surveillance data. These data are composed of reported cases of and/or positive laboratory test results for chlamydia, gonorrhea, and syphilis among San Francisco residents, which laboratories and clinicians are mandated by law to report.[15,16] The surveillance data are integrated with other data sources to comprise the SFDPH STI registry, which includes information on the individual's gender identity, reported gender identity of sex partners, and anatomic site of infection. The SFDPH STI registry also includes all chlamydia, gonorrhea, and syphilis test results for specimens tested through the San Francisco Public Health Laboratory. Data for analyses were accessed February 5, 2024. To provide context for doxyPEP use during the period of analysis, we separately describe doxyPEP prescribing by San Francisco clinicians after the release of citywide guidelines using sentinel surveillance data from the San Francisco Getting to Zero Consortium's New Project Initiative (eMethods in Supplement 1).

Study Design

We conducted an interrupted time series analysis to assess the ecologic association between the release and early implementation of San Francsico's doxyPEP guidelines and citywide reported cases of chlamydia, gonorrhea, and early syphilis among

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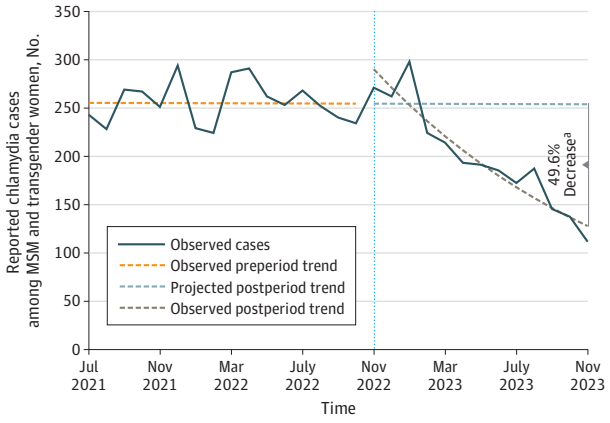
Figure 1. San Francisco Chlamydia Cases Among Men Who Have Sex With Men (MSM) and Transgender Women Before and After Doxycycline Postexposure Prophylaxis Guideline Implementation



The dotted vertical line represents the divide between the preperiod and postperiod after doxycycline postexposure prophylaxis guideline release. The solid dark blue line shows observed monthly case counts. The 3 dashed lines represent regression lines indicating the (1) observed preperiod trend, (2) observed postperiod trend, and (3) model-projected postperiod trend based on the preperiod trend.

[a]49.64% (95% CI, −59.05% to −38.06%) decrease in observed compared with projected chlamydia cases among MSM and transgender women in November 2023.

MSM and transgender women in San Francisco. Citywide doxyPEP guidance was released by SFDPH in October 2022. We defined the preimplementation and postimplementation periods as July 2021 through October 2022 and November 2022 through November 2023, respectively. A preperiod start date of July 2021 was selected because that is when the COVID-19 vaccine became widely available and access to health care services and STI screening had largely returned to pre-COVID-19 pandemic levels in San Francisco. This analysis was repeated for chlamydia among cisgender women, a population not recommended for doxyPEP use in the SFDPH guidance released in October 2022, to assess whether declines seen among MSM and transgender women could be due to other factors causing declines more broadly in San Francisco.

To account for potential delays between the release of guidelines and implementation of doxyPEP by clinicians and to minimize inclusion of STI screening results prior to a patient's first use of doxyPEP, we conducted a sensitivity analysis for cases among MSM and transgender women in which the start of the postimplementation period was shifted 3 months later to February 2023. To assess whether changes in reported STI cases could be attributed to changes in STI screening, we describe the mean number of chlamydia and gonorrhea tests per month and the mean number of syphilis screening tests per month conducted by the San Francisco Public Health Laboratory in the preimplementation and postimplementation periods.

Statistical Analysis

We calculated the number of monthly reported cases of chlamydia, gonorrhea, and early syphilis among MSM and transgender women during the period of interest. Cisgender men were classified as MSM if the SFDPH STI registry had a record of them having a reported cisgender male sex partner or a rectal or pharyngeal STI test in the year prior to infection. We fitted a segmented regression model using autoregressive integrated moving average errors to the log-transformed monthly case counts.[17] The parameters of the model allow for slopes before and after an intervention and for a change immediately following an intervention. Allowing for a change immediately following an intervention enables us to distinguish between short-term and longer-term trends around the time of doxyPEP implementation. The models yielded projected monthly case counts in the absence of doxyPEP implementation. The percentage change in observed compared with projected case counts at the end of the period of analysis (November 2023) were calculated. Analyses were conducted using R version 4.3.1 (R Core Team, 2023) with the forecast and the asta packages.

Results

Citywide Reported STI Cases Before and After DoxyPEP Guideline Release

Citywide, there were 6694 cases of chlamydia, 9603 cases of gonorrhea, and 2121 cases of early syphilis among MSM and transgender women during the analytic period. For chlamydia (**Figure 1**), monthly case counts among MSM and transgender women in the preimplementation period were flat (−0.02% per month; 95% CI, −0.91% to 0.88%). After doxyPEP implementation, there was an immediate 14.03% (95% CI, 0.90% to 28.86%) increase in cases followed by a sustained decrease of −6.58% per month (95% CI, −7.99% to −5.16%) compared with projections. By November 2023 (13 months after guideline release), chlamydia cases were −49.64% (95% CI, −59.05% to −38.06%) lower than projected based on preperiod trends. The sensitivity analysis of chlamydia cases (by shifting the postperiod to start in February 2023 instead of November 2022) among MSM and transgender women showed similar results, with a significant postperiod slope of −6.74% per month (95% CI, −8.47% to −4.97%) and a −62.30% (95% CI, −69.34% to −53.65%) decrease in observed cases compared with projected cases in November 2023. Among cisgender women, chlamydia cases in the postperiod increased significantly (2.46% per month; 95% CI, 0.77% to 4.21%) compared with projections (**Table**).

Early syphilis case counts in MSM and transgender women (**Figure 2**) were generally flat in the preperiod (0.15% per month; 95% CI, −0.52% to 0.82%). After doxyPEP implementation, there was an immediate −32.65% (95% CI, −38.74% to −25.95%) decrease in cases followed by an ongoing less pronounced decrease in cases by −2.68% per month (95% CI, −3.75% to −1.60%) compared with projections. In November 2023, the end of the 13-month postperiod, early syphilis cases decreased −51.39% (95% CI, −58.21% to −43.46%) compared with projected counts. The sensitivity analysis of early syphilis cases among MSM and transgender women showed that varying the postperiod to a later time

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Table. Interrupted Time Series Autoregressive Integrated Moving Average Modeling Analysis of Monthly San Francisco Sexually Transmitted Infection Cases Before and After Doxycycline Postexposure Prophylaxis Guideline Implementation

Population[a]	Preperiod slope, % change per mo (95% CI)[b]	Change October to November 2022, % (95% CI)[c]	Postperiod slope, % change per mo (95% CI)[d]	November 2023 change from projected, % (95% CI)
Chlamydia among MSM and transgender women				
Main	−0.02 (−0.91 to 0.88)	14.03 (0.90 to 28.86)	−6.58 (−7.99 to −5.16)	−49.64 (−59.05 to −38.06)
Sensitivity	0.35 (−0.32 to 1.02)	−12.91 (−22.80 to −1.75)	−6.74 (−8.47 to −4.97)	−62.30 (−69.34 to −53.65)
Early syphilis among MSM and transgender women				
Main	0.15 (−0.52 to 0.82)	−32.65 (−38.74 to −25.95)	−2.68 (−3.75 to −1.60)	−51.39 (−58.21 to −43.46)
Sensitivity	−1.66 (−2.91 to −0.39)	−17.18 (−34.29 to 4.38)	−1.50 (−4.98 to 2.11)	−30.90 (−53.52 to 2.72)
Gonorrhea among MSM and transgender women				
Main	−1.80 (−2.32 to −1.27)	1.75 (−5.69 to 9.77)	1.77 (0.87 to 2.67)	25.55 (−0.40 to 58.28)
Sensitivity	−1.36 (−2.01 to −0.70)	−3.66 (−14.51 to 8.57)	2.18 (0.30 to 4.10)	24.85 (1.72 to 53.23)
Chlamydia among cisgender women				
Main	−0.77 (−1.76 to 0.23)	−4.93 (−17.09 to 9.01)	2.46 (0.74 to 4.21)	27.29 (0.98 to 60.47)

Abbreviation: MSM, men who have sex with men.

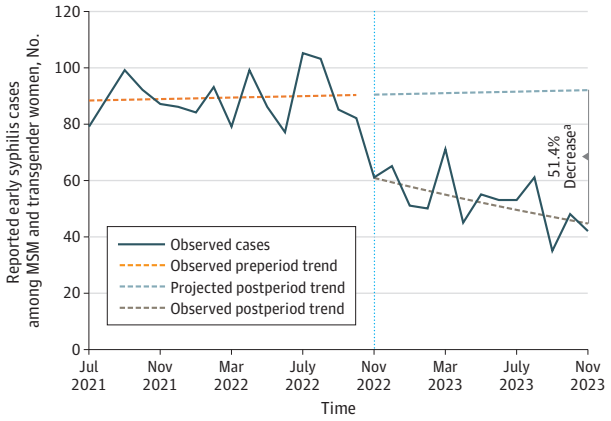
[a] The main analyses compare a preperiod of July 2021 through October 2022 to a postperiod of November 2022 through November 2023. The sensitivity analyses shift the intervention time point 3 months later, comparing a preperiod of July 2021 through January 2023 to a postperiod of February 2023 through November 2023.

[b] Preperiod slope represents percentage change in case counts per month during the preperiod.

[c] Change from October to November 2022 represents a 1-time percentage change in case counts from the last value during the preperiod to the first projected value in the postperiod.

[d] Postperiod slope represents the percentage change in case counts per month during the postperiod comparing model projections with observed counts.

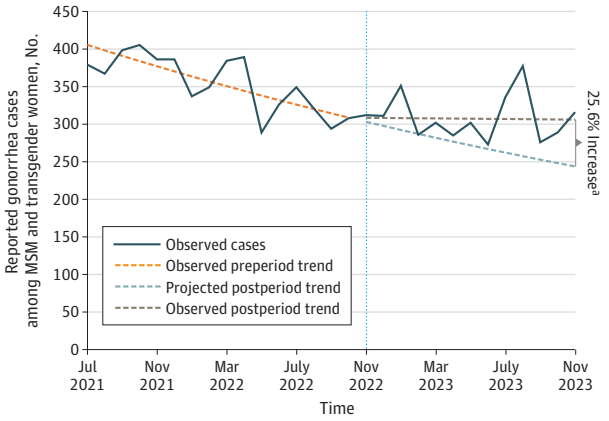
Figure 2. San Francisco Early Syphilis Cases Among Men Who Have Sex With Men (MSM) and Transgender Women Before and After Doxycycline Postexposure Prophylaxis Guideline Implementation



The dotted vertical line represents the divide between the preperiod and postperiod after doxycycline postexposure prophylaxis guideline release. The solid dark blue line shows observed monthly case counts. The 3 dashed lines represent regression lines indicating the (1) observed preperiod trend, (2) observed postperiod trend, and (3) model-projected postperiod trend based on the preperiod trend.

[a] 51.39% (95% CI, −58.21% to −43.46%) decrease in observed compared with projected early syphilis cases among MSM and transgender women in November 2023.

Figure 3. San Francisco Gonorrhea Cases Among Men Who Have Sex With Men (MSM) and Transgender Women Before and After Doxycycline Postexposure Prophylaxis Guideline Implementation



The dotted vertical line represents the divide between the preperiod and postperiod after doxycycline postexposure prophylaxis guideline release. The solid dark blue line shows observed monthly case counts. The 3 dashed lines represent regression lines indicating the (1) observed preperiod trend, (2) observed postperiod trend, and (3) model-projected postperiod trend based on the preperiod trend.

[a] 25.55% (95% CI, −0.40% to 58.28%) increase in observed compared with projected gonorrhea cases among MSM and transgender women in November 2023.

attenuated the effects attributed to doxyPEP implementation to a −30.90% (95% CI, −53.52% to 2.72%) decrease in November 2023 (Table).

For gonorrhea, preperiod cases among MSM and transgender women were declining by −1.80% per month (95% CI, −2.32% to −1.27%). This trend stopped just as the doxyPEP guidelines were released, and the number of gonorrhea cases per month stabilized (**Figure 3**). Because of this stabilization, compared with the preperiod, there was a net increase in gonorrhea cases of 1.77% per month (95% CI, 0.87% to 2.67%); by the end of the 13-month postperiod, gonorrhea cases were higher by 25.55% (95% CI, −0.40% to 58.28%) compared with what would have been expected if the preperiod trend had continued. The sensitivity analysis of gonorrhea cases among MSM and transgender women showed similar results, with an ongoing increase of 2.18% per month (95% CI, 0.30% to 4.10%)

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compared with projected counts and a 24.85% (95% CI, 1.72% to 53.23%) increase in observed cases compared with projected counts in November 2023 (Table).

San Francisco Public Health Laboratory Testing Volume

There were a mean (SD) 6527 (439) chlamydia tests per month performed at the San Francisco Public Health Laboratory in the preperiod compared with 6597 (482) tests per month in the postperiod. For syphilis and gonorrhea, there were a mean (SD) 1764 (234) and 6523 (438) tests per month, respectively, in the preperiod compared with 1733 (180) and 6598 (484) tests per month in the postperiod.

DoxyPEP Uptake at Sentinel Clinics

From November 2022 to December 2023, 20 406 MSM and transgender women were seen at the 3 clinics where doxyPEP uptake is being monitored, and 3974 (19.5%) initiated doxyPEP (eResults and eTable in Supplement 1). DoxyPEP initiations began immediately following guidance release and continued through the period analyzed, with 583 starts in 2022 quarter 4, 1311 in 2023 quarter 1, 816 in 2023 quarter 2, 731 in 2023 quarter 3, and 533 in 2023 quarter 4.

Discussion

There have been substantial and sustained declines in reported cases of chlamydia and early syphilis, but not gonorrhea, among MSM and transgender women in San Francisco in the 13 months following the release and implementation of SFDPH doxyPEP guidelines. While we cannot definitively rule out that these declines were due to or affected by other factors, for example changes in sexual behaviors or STI screening practices, the lack of decline in chlamydia in cisgender women (for whom doxyPEP was not recommended) and the abrupt change in trajectory in case counts after a decade of rising STI rates in MSM suggest that these declines were due at least in part to doxyPEP use.

Despite the impact of doxyPEP use on gonorrhea incidence in randomized clinical trials, in this analysis, doxyPEP implementation was not associated with a significant decrease in citywide cases of gonorrhea in MSM and transgender women. There are several possible reasons for the lack of association. While doxyPEP significantly reduced gonorrhea incidence in MSM and transgender women enrolled in the doxyPEP randomized trials, it was less effective against gonorrhea compared with chlamydia or syphilis. It is possible that doxyPEP adherence or use patterns in randomized trial setting are different than in the community. For example, taking doxyPEP soon after exposure and after every sexual encounter may be more important for the prevention of gonorrhea than for chlamydia or early syphilis, as doxycycline concentrations in the rectum and urethra are closer to the minimum inhibitory concentration (MIC) threshold for gonorrhea and remain above the MIC for gonorrhea for a shorter duration of time compared with chlamydia or syphilis.[18] Second, it is possible that the prevalence of tetracycline-resistant gonorrhea strains has increased in San Francisco. DoxyPEP may be less effective at higher tetracycline MICs; however, the threshold MIC for doxyPEP efficacy is not known. Doxycycline use had been increasing in San Francisco in sexual health care settings prior to release of the doxyPEP guidelines due to a shift in 2021 in first-line recommended treatment for chlamydia infection and urethritis from azithromycin to doxycycline.[19,20] In England, where doxyPEP has not yet been formally recommended, there is evidence of increasing tetracycline-resistant in gonorrhea.[21] Monitoring gonorrhea antibiotic susceptibility both overall and in individuals using doxyPEP will be important as implementation continues to scale-up. Additionally, conducting surveillance for antimicrobial resistance in non-STI bystander bacteria, including *Staphylococcus aureus*, *Streptococcus pneumoniae*, and *Escherichia coli*, will be important to assess for other potential adverse sequelae of doxyPEP use.

Centers for Disease Control and Prevention doxyPEP guidelines state that evidence is insufficient to assess the balance of benefits and harms of the use of doxyPEP for cisgender women. While rates of chlamydia and early syphilis declined among MSM and transgender women during the time frame of this analysis, there was an increase in chlamydia and early syphilis in cisgender women in San Francisco. Additional studies of doxyPEP and other strategies for STI prevention in cisgender women and cisgender men who have sex with cisgender women are urgently needed.

Limitations

This analysis has limitations. First, this is an ecologic analysis, and we cannot conclude a causal relationship between release of doxyPEP guidelines and STI trends in San Francisco. Second, we have limited data about doxyPEP prescribing, adherence, and use patterns in San Francisco during this period. The proportion of a population at risk for STIs that would need to use doxyPEP to impact population-level STI rates is unknown. Network-centric STI prevention interventions that reach central nodes in highly interconnected networks may have significant impacts on transmission of infections. It is possible that MSM and transgender women who have elected to use doxyPEP in San Francisco have a higher risk of STIs than others, thus enabling a population-level reduction in STIs without broad use.[22] Results from the sentinel surveillance clinics show that interest in doxyPEP was high and clinics began prescribing it immediately after guideline release. The sentinel surveillance data are an underestimate of citywide doxyPEP use and do not include the more than 1200 individuals prescribed doxyPEP at a large health maintenance organization in San Francisco by December 2023 (Jonathan Volk, MD, Kaiser Permanente San Francisco, personal communication, May 28, 2024), nor individuals prescribed doxyPEP by private or safety-net primary care clinicians. The number of MSM and transgender women in San Francisco who had used doxyPEP prior to October 2022 is unknown. There were 208 individuals prescribed doxyPEP as part of the DoxyPEP randomized trial between September 2020 and May 2022 and likely others who accessed doxycycline from their health care clinician or through other means. While it is possible that use of doxycycline after sexual encounters prior to the release of guidelines could have impacted preperiod STI trends, we expect this impact to be

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minimal. If there were an impact, it would mean the declines observed for chlamydia and early syphilis would be an underestimate of what would have been seen in the absence of any preguideline use. Third, the time frame of the analysis followed a dynamic period during which effects of COVID-19 or an outbreak of mpox in summer 2022 may have had lingering effects on STI screening rates, sexual practices, and condom use, making it difficult to disentangle the impact of doxyPEP (eFigure in Supplement 1). Conducting this analysis in jurisdictions with a similarly high incidence of STIs but low doxyPEP rollout could strengthen the findings that declining chlamydia and early syphilis in San Francisco is related to doxyPEP use. However, the concurrent increase in chlamydia cases in cisgender women in San Francisco, a population for whom doxyPEP use is rare, supports the hypothesis that reductions in chlamydia and early syphilis seen among MSM and transgender women were associated with doxyPEP uptake and not only with other factors causing reductions in STIs in San Francisco more broadly.[23]

Fourth, because clinicians and laboratories are not mandated to report negative STI tests, we are unable to assess whether declines seen in chlamydia and early syphilis were due to decreases in testing citywide. However, there was little change in the average number of chlamydia, gonorrhea, or syphilis tests per month conducted at San Francisco Public Health Laboratory, the site of STI testing for more than 50 clinics in San Francisco, when comparing the preperiod and postperiod, suggesting that there were not marked decreases in testing citywide.

Fifth, the early and abrupt decline in early syphilis in the fourth quarter of 2022 is unlikely to be explained by doxyPEP alone. In fact, our sensitivity analysis found that if we shifted the beginning of the postperiod to 3 months later, there was no longer a significant difference in projected and observed early syphilis case counts in November 2023; however, the decline in syphilis was sustained and the trend toward lower observed case counts persisted. Lastly, San Francisco is a unique city with a large population of MSM and transgender individuals; the sexual network structure and thus the impacts of a prevention tool may differ in jurisdictions with more disparate or less interconnected networks.[24]

Conclusions

This analysis demonstrates the potential for a substantial population-level impact of doxyPEP on STI incidence among MSM and transgender women. Future analyses are planned with extended postperiod data to determine whether observed trends continue to align with citywide doxyPEP uptake and to assess for demographic disparities in doxyPEP uptake and STI incidence. Additional data sources to characterize doxyPEP prescriptions and adherence more broadly are needed. Meanwhile, supporting doxyPEP implementation through clinician and patient education and ensuring low-barrier access to doxyPEP for MSM and transgender women with a greater likelihood of STIs could have a significant impact on the nationwide epidemic of STIs.

ARTICLE INFORMATION

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Author Contributions: Ms Sankaran had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Sankaran, Glidden, Bacon, Buchbinder, Gandhi, Havlir, Cohen.
Acquisition, analysis, or interpretation of data: Sankaran, Glidden, Kohn, T. Nguyen, Buchbinder, Gandhi, Havlir, Liebi, Luetkemeyer, J. Nguyen, Roman, Scott, Torres, Cohen.
Drafting of the manuscript: Sankaran, Glidden, Luetkemeyer, Cohen.
Critical review of the manuscript for important intellectual content: Sankaran, Kohn, T. Nguyen, Bacon, Buchbinder, Gandhi, Havlir, Liebi, Luetkemeyer, J. Nguyen, Roman, Scott, Torres, Cohen.
Statistical analysis: Sankaran, Glidden, Kohn, J. Nguyen, Torres.
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Data Sharing Statement: See Supplement 2.

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